

1. Theory of Interest

Accumulation Functions

Simple: $a(t) = 1 + rt$ Compound: $a(t) = (1 + r)^t$ [default]

Continuous: $a(t) = e^{\delta t}$

Accumulated value of principal P : $A(t) = P \cdot a(t)$

Frequency of Compounding

Nominal rate $r^{(p)}$ paid p times/yr:

$$a(t) = \left(1 + \frac{r^{(p)}}{p}\right)^{pt}$$

Effective annual rate:

$$r_e = \left(1 + \frac{r^{(p)}}{p}\right)^p - 1$$

Equivalence of two nominal rates:

$$\left(1 + \frac{r^{(p)}}{p}\right)^p = \left(1 + \frac{R^{(q)}}{q}\right)^q$$

$r_e > r^{(p)}$ for any $p > 1$. Continuous compounding maximises r_e for a given nominal rate.

Force of Interest

$$\delta(t) = \frac{a'(t)}{a(t)} = \frac{d}{dt} \ln a(t)$$

Constant δ : $a(t) = e^{\delta t}$, $\delta = \ln(1 + r_e)$, $r_e = e^{\delta} - 1$.

Variable force of interest:

$$a(t) = \exp\left(\int_0^t \delta(u) du\right) \quad a(s, t) = \exp\left(\int_s^t \delta(u) du\right)$$

Key Conversions (quick ref)

Convert	Formula
$r^{(p)} \rightarrow r_e$	$(1 + r^{(p)}/p)^p - 1$
$r_e \rightarrow r^{(p)}$	$p[(1 + r_e)^{1/p} - 1]$
$r_e \rightarrow \delta$	$\ln(1 + r_e)$
$\delta \rightarrow r_e$	$e^{\delta} - 1$
$r^{(p)} \rightarrow r^{(q)}$	equate $(1 + r^{(p)}/p)^p = (1 + r^{(q)}/q)^q$

2. Present Value & Cash Flows

Present Value (single payment)

$$PV = \frac{c}{a(t)} = \frac{c}{(1 + r)^t}$$

Discount factor: $v = \frac{1}{1+r}$, so $PV = c \cdot v^t$

PV of Cash Flow Stream

$C = \{(c_i, t_i)\}_{i=1}^n$:

$$PV(C) = \sum_{i=1}^n \frac{c_i}{(1 + r)^{t_i}}$$

Time value at t : $TV_t(C) = PV(C) \cdot (1 + r)^t$

Annuities

Let r = effective rate per period, A = payment per period, n = number of periods.

Ordinary annuity (payment at end of each period):

$$PV = \frac{A}{r} \left[1 - \frac{1}{(1 + r)^n}\right]$$

Annuity-due (payment at start of each period):

$$PV = \frac{A(1 + r)}{r} \left[1 - \frac{1}{(1 + r)^n}\right]$$

$\Rightarrow PV(\text{due}) = PV(\text{ordinary}) \times (1 + r)$

Perpetuity (ordinary, payments at end):

$$PV = \frac{A}{r}$$

Perpetuity-due (payments at start):

$$PV = \frac{A(1 + r)}{r}$$

Future value (ordinary annuity at $t = n$):

$$FV = \frac{A}{r} [(1 + r)^n - 1]$$

Deferred annuity (n payments, first at $t = m + 1$):

$$PV = \frac{A}{r} \left[1 - \frac{1}{(1 + r)^n}\right] \cdot \frac{1}{(1 + r)^m}$$

Discount ordinary annuity PV back by m periods.

Rate mismatch: Always convert nominal rate to effective rate per payment period before applying annuity formula. E.g. 6% p.a. convertible semi-annually $\rightarrow$ effective monthly rate $= (1 + 0.03)^{1/6} - 1$.

NPV & IRR Decision Rules

NPV criterion: Invest if $NPV > 0$. Prefer A over B if $NPV(A) > NPV(B)$.

IRR criterion: Invest if $IRR >$ required return. Prefer A over B if $IRR(A) > IRR(B)$.

IRR is r s.t. $PV(C) = 0$. Uniqueness: if $c_0 < 0$, $c_k \geq 0 \forall k \geq 1$, $\sum c_k > 0 \Rightarrow$ unique IRR.

Newton–Raphson: $r_{n+1} = r_n - f(r_n)/f'(r_n)$

Growing Annuity & Perpetuity

First payment A at $t = 1$, grows at rate g /period, n periods:

$$PV = \frac{A}{r - g} \left[1 - \left(\frac{1 + g}{1 + r}\right)^n\right] \quad (r \neq g) \quad r = g : \frac{nA}{1 + r}$$

Growing perpetuity ($r > g$): $PV = A/(r - g)$.

Loans, Refinancing & Last Payment

Loan L , n payments A , rate r /period: $L = \frac{A}{r} [1 - (1 + r)^{-n}]$,

$$A = \frac{Lr}{1 - (1 + r)^{-n}}$$

Balance after m payments: $L_m^{\text{bal}} = \frac{A}{r} [1 - (1 + r)^{-(n-m)}]$ (prospective)

$L_m^{\text{bal}} = L(1 + r)^m - \frac{A}{r} [(1 + r)^m - 1]$ (retrospective)

Irregular last payment (n non-integer): make $[n]$ full payments of A plus final smaller payment B .

• B at $t = [n]$: set $L = PV$ of $[n] - 1$ payments of A plus B at $t = [n]$.

• B at $t = [n] + 1$: use $L_{[n]}^{\text{bal}}$, then $B = L_{[n]}^{\text{bal}}(1 + r)$.

3. Bonds

Notation

F face value (= R unless stated)
 R redemption value
 c nominal coupon rate (annual)
 m coupon payments per year
 n total coupon payments
 λ nominal bond yield (annual)
 P bond price

Bond Pricing Formula

$$P = \frac{R}{(1 + \lambda/m)^n} + \sum_{i=1}^n \frac{cF/m}{(1 + \lambda/m)^i}$$

$$= F \left[\frac{1}{(1 + \lambda/m)^n} + \frac{c/m}{\lambda/m} \left(1 - \frac{1}{(1 + \lambda/m)^n}\right) \right] \quad (F = R)$$

Premium/discount form ($F = R$): $P = F + F \cdot \frac{c - \lambda}{\lambda} [1 - (1 + \lambda/m)^{-n}]$

$c > \lambda \Rightarrow P > F$ (premium); $c = \lambda \Rightarrow P = F$ (par); $c < \lambda \Rightarrow P < F$ (discount).

Makeham formula (alternative; let $K = R/(1 + \lambda/m)^n$):

$$P = K + \frac{c}{\lambda} (F - K) \quad (F = R \text{ case})$$

Recursive price: $P_{k+1} = P_k(1 + \lambda/m) - cF/m$ (price after k -th coupon).

Variable Coupon Bonds

Split into groups with constant coupon rate; price each group as a separate annuity + shared redemption PV.

Exam pattern (Q3 midterm): Bond with different coupon rates for each 5-year block and $R \neq F$. Price each block as an annuity discounted at yield λ , then add PV of redemption at maturity.

Determining Bond Yield (Newton–Raphson)

$f(\lambda) = P(\lambda) - P_{\text{market}}$, iterate $\lambda_{n+1} = \lambda_n - f(\lambda_n)/f'(\lambda_n)$.

$$P(\lambda) = \frac{R}{(1 + \lambda/m)^n} + \frac{cF/m}{\lambda/m} \left[1 - \frac{1}{(1 + \lambda/m)^n}\right]$$

$f'(\lambda)$ for annual coupon bond ($m = 1$, $F = R$):

$$f'(\lambda) = - \sum_{i=1}^n \frac{i \cdot cF}{(1 + \lambda)^{i+1}} - \frac{nF}{(1 + \lambda)^{n+1}} = - \frac{D \cdot P(\lambda)}{1 + \lambda}$$

where D = Macaulay duration. (In general: $f'(\lambda) = - \frac{D \cdot P}{m(1 + \lambda/m)} \cdot$)

Start: $\lambda_0 = c$. Price–yield: **inverse** & **convex** ($P \downarrow$ as $\lambda \uparrow$).

Macaulay Duration

$$D = \frac{\sum_{i=1}^n t_i \cdot PV(c_i, t_i)}{PV(C)} = \sum_{i=1}^n w_i t_i, \quad w_i = \frac{PV(c_i)}{PV(C)}$$

Special cases ($\mu = \lambda/m$, $\gamma = c/m$): Zero-coupon: $D = T$. Perpetuity : $D = (1 + \lambda/m)/\lambda$.

Coupon bond, $F = R$ (general):
$$D = \frac{1 + \mu}{m\mu} - \frac{1 + \mu + n(\gamma - \mu)}{m\gamma[(1 + \mu)^n - 1] + m\mu} \quad (2)$$

At par $P = F = R$ ($\mu = \gamma$):
$$D = \frac{1 + \mu}{m\mu} \left[1 - \frac{1}{(1 + \mu)^n}\right] \quad (3)$$

$t_1 \leq D \leq t_n$; D independent of F and of yield compounding frequency.

4. Term Structure

Spot Rates & Forward Rates

Spot rate s_t : effective annual rate from 0 to t . Accumulation: $(1 + s_t)^t$.

Forward rate $f_{j,k}$: rate from $t = j$ to $t = k$ (annual, same convention as spot).

No-arbitrage relation:

$$(1 + s_k)^k = (1 + s_j)^j (1 + f_{j,k})^{k-j}$$

Note: $f_{0,t} = s_t$.

Continuous compounding version:

$$e^{s_k k} = e^{s_j j} \cdot e^{f_{j,k}(k-j)} \Rightarrow f_{j,k} = \frac{s_k k - s_j j}{k - j}$$

Bootstrapping Spot Rates

Use observed coupon bond prices to extract $s_1, s_2, \dots$ sequentially:

For a coupon bond with annual payments:

$$P = \frac{c_1}{1 + s_1} + \frac{c_2}{(1 + s_2)^2} + \dots + \frac{c_n + F}{(1 + s_n)^n}$$

Solve for the unknown s_n given $s_1, \dots, s_{n-1}$ already found.

Exam pattern: Extract s_1, s_2 from zero-coupon bond prices, then use coupon bond price to find s_3 , then compute $f_{1,3}$.

Yield Curve Shapes

Normal (upward): economic expansion, low inflation.
Inverted (downward): pre-recession, high inflation.

Flat term structure: all s_t equal $\Rightarrow$ force of interest is constant.

5. Asset Returns & Portfolios

Returns

Total return: $R = P_1/P_0$ Rate of return: $r = (P_1 - P_0)/P_0$ $R = 1 + r$

Short sell: $R_{\text{short}} = P_1/P_0$ (same formula – negatives cancel).

Definitions & Identities

$$\text{Var}(X) = E[X^2] - E[X]^2 \quad \text{Cov}(X, Y) = E[XY] - E[X]E[Y] \quad \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Expectation (linearity): $E(c) = c$; $E(cX) = cE(X)$; $E(X + c) = E(X) + c$; $E(aX + bY) = aE(X) + bE(Y)$

Variance: $\text{Var}(aX + b) = a^2 \text{Var}(X)$; $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2\text{Cov}(X, Y)$

Covariance: $\text{Cov}(X, Y) = \text{Cov}(Y, X)$; $\text{Cov}(X, X) = \text{Var}(X)$; $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$

Bilinearity: $\text{Cov}(aX + bY, Z) = a \text{Cov}(X, Z) + b \text{Cov}(Y, Z)$
Indep $\Rightarrow \text{Cov} = 0$; $\text{Cov} = 0 \nRightarrow$ indep.

Asset Returns (notation)

Mean: $\mu_i = E(r_i)$; Var: σ_i^2 ; Cov: $\sigma_{ij} = \text{Cov}(r_i, r_j)$

Short-sell: profit $= P_0 - P_1$, return $= -r_{\text{long}}$, loss unlimited.

Portfolio of 2 Assets

Weights $w_1 = \alpha$, $w_2 = 1 - \alpha$ ($\alpha \in \mathbb{R}$ if short-selling allowed; $\alpha \in [0, 1]$ otherwise).

$$\mu_p = \alpha \mu_1 + (1 - \alpha) \mu_2$$

$$\sigma_p^2 = \alpha^2 \sigma_1^2 + (1 - \alpha)^2 \sigma_2^2 + 2\alpha(1 - \alpha) \rho_{12} \sigma_1 \sigma_2$$

Global Minimum-Variance Portfolio (GMVP)

$$\alpha^* = \frac{\sigma_2^2 - \rho_{12}\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$$

$$(\sigma_p^2)^* = \frac{\sigma_1^2\sigma_2^2(1 - \rho_{12}^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$$

If $\rho = 1$: feasible set is a straight line. If $\rho = -1$: V-shape (zero variance achievable). If $-1 < \rho < 1$: curved bullet shape.

Efficient frontier: upper half of feasible set (including GMVP).

Check: if $\alpha^* \notin [0, 1]$ and short-selling not allowed, the GMVP is the asset with lower σ .

6. Forwards & Futures

Forward Contract

Obligation to buy/sell at forward price $F(t, T)$ at maturity T .

- Long payoff: $S_T - F(t, T)$
- Short payoff: $F(t, T) - S_T$
- Costs nothing to enter (no premium)

Forward Pricing (No-Arbitrage)

Non-dividend-paying asset:

$F(0, T) = S_0e^{rT} \text{ (cts)} \quad F(0, T) = S_0(1 + r)^T \text{ (disc)}$
General t: $F(t, T) = S_t e^{r(T-t)}$ With storage cost c (lump sum at $t = 0$): $F(0, T) = (S_0 + c) e^{rT}$ With continuous dividend yield q: $F(0, T) = S_0 e^{(r-q)T}$
Forward price does not depend on expected return μ of the underlying — only on S_0, r, T (and carry costs). This is a no-arbitrage result, not an expectation.

Arbitrage Construction

If $F(0, T) > S_0e^{rT}$: short forward, buy asset (borrow S_0). Profit at T : $F - S_0e^{rT} > 0$.
If $F(0, T) < S_0e^{rT}$: long forward, short-sell asset (invest proceeds). Profit at T : $S_0e^{rT} - F > 0$.

Futures vs. Forwards

	Forward	Futures
Settlement	At maturity	Daily MTM
Counterparty	Bilateral	Exchange

Daily P&L long futures: $\approx F_{t+1} - F_t$ (margin account).

Margin Account

Initial margin deposited. If balance falls below **maintenance margin**, margin call issued (top up to initial margin).

7. Hedging

Perfect Hedge (Forwards/Futures)

Enter opposite position to fix price and eliminate risk. Conditions for perfect futures hedge: (1) delivery date matches; (2) same underlying; (3) integer multiple of contract size; (4) sufficient liquidity.

Hedging Direction ($\rho > 0$)

Obligation	Futures action	
Buy asset A	Short futures B	If $\rho < 0$: reverse direction.
Sell asset A	Long futures B	

Minimum-Variance Hedge

Hedge W units of asset A using futures on asset B (correlated). Short h futures contracts.

Cash flow at T : $y = -WS_T + (F_T - F_0)h$

$\text{Var}(y) = W^2\text{Var}(S_T) - 2WCov(S_T, F_T)h + \text{Var}(F_T)h^2$
Optimal hedge ratio: $h^* = W \cdot \frac{\text{Cov}(S_T, F_T)}{\text{Var}(F_T)} = W \cdot \frac{\rho_{S,F} \sigma_S}{\sigma_F}$
Minimum variance: $\text{Var}^*(y) = W^2\text{Var}(S_T)(1 - \rho_{S,F}^2)$
Hedge effectiveness: $\rho_{S,F}^2$. Factor $\sqrt{1 - \rho^2}$ measures residual risk. h^* is in <i>units</i>, not contracts. Number of contracts = $h^*/(\text{contract size})$. Round to integer for real problems.

8. Options

Key Definitions

Call: right to *buy* at strike K . **Put**: right to *sell* at strike K .

Long = holder (pays premium); Short = writer (receives premium).

European: exercise only at T . American: exercise any time $\leq T$. (Assume European unless stated.)

Payoffs & Profits

Position	Payoff at T	Profit
Long call	$\max(S_T - K, 0)$	payoff $-Ce^{rT}$
Short call	$-\max(S_T - K, 0)$	payoff $+Ce^{rT}$
Long put	$\max(K - S_T, 0)$	payoff $-Pe^{rT}$
Short put	$-\max(K - S_T, 0)$	payoff $+Pe^{rT}$

C, P = option premium (time-0 price). Time-value of premium at T : Ce^{rT} or Pe^{rT} .

Bounds

Upper bounds: $C \leq S_0 \quad P \leq Ke^{-rT}$

Lower bounds:

$C \geq \max(0, S_0 - Ke^{-rT}) \quad P \geq \max(0, Ke^{-rT} - S_0)$

Bounds Arbitrage

$C > S_0$: short call, long stock.

	$t=0$	$S_T > K$	$S_T \leq K$
Short call	$+C$	$-(S_T - K)$	0
Long stock	$-S_0$	$+S_T$	$+S_T$
Total	$C - S_0 > 0$	$+K > 0$	$S_T \geq 0$

$P > Ke^{-rT}$: short put, invest Ke^{-rT} .

	$t=0$	$S_T < K$	$S_T \geq K$
Short put	$+P$	$-(K - S_T)$	0
Bond	$-Ke^{-rT}$	$+K$	$+K$
Total	$P - Ke^{-rT} > 0$	$S_T \geq 0$	$+K > 0$

$C < S_0 - Ke^{-rT}$: short stock, long call, invest $S_0 - C$
 $[I \equiv (S_0 - C)e^{rT}]$.

	$t=0$	$S_T \geq K$	$S_T < K$
Short stock	$+S_0$	$-S_T$	$-S_T$
Long call	$-C$	$+(S_T - K)$	0
Invest	$-(S_0 - C)$	$+I$	$+I$
Total	0	$I - K > 0$	$I - S_T > 0$

$P < Ke^{-rT} - S_0$: long put, long stock, borrow $S_0 + P$
 $[B \equiv (S_0 + P)e^{rT}]$.

	$t=0$	$S_T < K$	$S_T \geq K$
Long put	$-P$	$+(K - S_T)$	0
Long stock	$-S_0$	$+S_T$	$+S_T$
Borrow	$+(S_0 + P)$	$-B$	$-B$
Total	0	$K - B > 0$	$S_T - B \geq 0$

Put–Call Parity (European, no dividends):

$C + Ke^{-rT} = P + S_0$			
		$t=T$ $S_T > K$	$S_T < K$
Port. A	Call ZC bond (K) <i>Total</i>	$S_T - K$ K S_T	0 K K
Port. C	Put Share <i>Total</i>	0 S_T S_T	$K - S_T$ S_T K
Same payoff $\Rightarrow PV_A = PV_C$: $C + dK = P + S_0$.			

Put–Call Parity (General Form)

Let d = discount factor to maturity. Slides notation:

$C + dK = P + S_0$
Discrete: $d = (1 + r)^{-T}$. Continuous: $d = e^{-rT}$. Rearrange to find arbitrage: if $C + dK \neq P + S_0$, one side is cheap.

Trading Strategies

Bull spread: Long C_{K_1} + Short C_{K_2} , $K_1 < K_2$. Capitalise on price increase, cap max profit to reduce cost & risk.

S_T range	Long C_{K_1}	Short C_{K_2}	Total
$S_T \leq K_1$	0	0	0
$K_1 < S_T < K_2$	$S_T - K_1$	0	$S_T - K_1$
$S_T \geq K_2$	$S_T - K_1$	$K_2 - S_T$	$K_2 - K_1$

Bear spread: Short C_{K_1} + Long C_{K_2} or Long P_{K_2} + Short P_{K_1} . Profit if $S_T \downarrow$.

Butterfly ($K_1 < K_2 < K_3$, $2K_2 = K_1 + K_3$): Long C_{K_1} , Short $2C_{K_2}$, Long C_{K_3} . Low-vol bet ($S_T \approx K_2$).

S_T range	C_{K_1}	C_{K_3}	$-2C_{K_2}$	Total
$S_T \leq K_1$	0	0	0	0
$K_1 < S_T \leq K_2$	$S_T - K_1$	0	0	$S_T - K_1$
$K_2 < S_T < K_3$	$S_T - K_1$	0	$-2(S_T - K_2)$	$K_3 - K_1$
$S_T \geq K_3$	$S_T - K_1$	$S_T - K_3$	$-2(S_T - K_2)$	0

Straddle: Long call + Long put (same K, T). Profit if $|S_T - K|$ large.

Covered call: Long stock + Short call. **Protective put:** Long stock + Long put.

Binomial Option Pricing

Single period: $S \rightarrow uS$ or dS , risk-free rate r , $R = 1 + r$. No-arb requires $u > R > d$.

Risk-neutral probability: $q = \frac{R - d}{u - d}$, $1 - q = \frac{u - R}{u - d}$

$$C = \frac{1}{R}[qC_u + (1 - q)C_d]$$

$C_u = \max(uS - K, 0)$, $C_d = \max(dS - K, 0)$ for call; $\max(K - \cdot, 0)$ for put.

Multi-period: backward induction. Each node same formula. **CRR:** $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$.

Replication Portfolio (Binomial)

Construct portfolio: x shares + b in risk-free cash. Payoffs:

$\text{up: } ux + Rb = C_u \quad \text{down: } dx + Rb = C_d$

Solve:

$$x = \frac{C_u - C_d}{u - d} \quad b = \frac{uC_d - dC_u}{R(u - d)}$$

$$C = x + b = \frac{qC_u + (1 - q)C_d}{R} \quad \checkmark$$

Arbitrage: if $C > x + b$: short call, long portfolio. If $C < x + b$: long call, short portfolio.

Black–Scholes (Intro Only)

Stock GBM: $dS = \mu S dt + \sigma S dz$. CRR $\rightarrow$ B–S as $\Delta t \rightarrow 0$: $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$.

Exam Patterns (Midterm)

Q1 type — PV algebra: Set up PV equation, let $v = (1 + r)^{-1}$ or $x = (1 + r)^{-2k}$, solve quadratic/polynomial. Answer in terms of k .

Q2 type — Loan refinancing: (i) Use annuity formula to find L/X (match L =PV of annuity). (ii) Find balance after m payments. Convert interest rate to *effective per period*. New annuity equation for Y . **Key trap:** Lump sum paid immediately, then first instalment one month later (ordinary annuity shifted by 1 month in timeline).

Q3 type — Variable-coupon bond + $R \neq F$: Split into 4 annuity blocks (10 semi-annual payments each at different rates). Write PV equation with P on both sides (since $R = P + 20$), solve linear equation.

Q4 type — Newton–Raphson for bond yield: Set $f(\lambda)$ = PV of cash flows at $\lambda - P_{\text{market}}$. Compute $f(0)$ as a starting check, choose λ_0 near expected answer, iterate 1–2 times for 4 d.p.

Common Mistakes

- Using nominal rate directly in annuity formula without converting
- Forgetting that annuity-due = ordinary annuity $\times (1 + r)$
- Coupon = cF/m (not cF) for semi-annual bonds
- Duration: weight by PV, not raw cash flow
- Forward price: uses *risk-free* rate r , not expected return μ
- Put–call parity: discount factor is e^{-rT} (continuous) or $(1 + r)^{-T}$ (discrete)